Volatility Forecasting

Extension to the eight-ETF portfolio study · out-of-sample, 2021-12-30 to 2025-12-31 · independent project

Question

Returns are hard to predict, but volatility clusters and is partly predictable. Which model forecasts next-day and next-month volatility best out of sample, and do better forecasts fix the under-coverage of 99% VaR found in the version-2 backtest?

Method

Ten return series (Main, Benchmark and the eight ETFs). Six models: historical variance over 63 and 252 sessions, EWMA (lambda 0.94), GARCH(1,1), GJR-GARCH(1,1) with a leverage term, and HAR on daily, weekly and monthly squared returns. GARCH-type parameters are estimated by Gaussian quasi-maximum likelihood (own Nelder-Mead optimiser) on an expanding window and re-estimated every 21 sessions; HAR is re-fitted on the same schedule using only targets already observed. Each forecast for day t uses data up to t-1. Losses: QLIKE (robust to the noisy squared-return proxy) and MSE; Diebold-Mariano tests against EWMA with Newey-West variance; Mincer-Zarnowitz regressions for bias. VaR: normal quantile versus filtered historical simulation (past standardized returns scaled by today's forecast).

Findings

• For Main, GJR-GARCH has the lowest QLIKE loss at both horizons, but does not beat EWMA significantly (Diebold-Mariano p = 0.09 at one day, 0.28 at one month).

• Across all 10 series, a long 252-session historical window is significantly worse than EWMA in 16 of 20 series-horizon tests ($p < 0.05$). Only 3 of 100 model-vs-EWMA tests show a significant improvement over EWMA: the big gain comes from reacting to recent volatility at all, not from the choice among conditional models.

• GJR-GARCH, which lets falls raise volatility more than rises, has the lowest loss in 6 of 10 series at one day and 5 at one month; its median loss is 0.99x EWMA (1 day) and 1.00x (1 month).

• Asset class matters: GJR-GARCH significantly beats EWMA only for Benchmark, VTI (US equity and the VTI-heavy benchmark), consistent with the leverage effect being strongest in equities; on bond ETFs the GARCH-type and HAR models are significantly worse than EWMA in 10 tests (BND, BNDX, IEF, TIP), so a simple EWMA is the safer default for fixed income.

• Fixing 99% VaR needs both a responsive volatility model and a fat-tailed distribution. Over 1005 days (expected 10 exceptions): the slow historical windows fail Kupiec's test even with filtered historical simulation (18 and 21 exceptions); the conditional models with a normal quantile fail in 2 of 4 cases; conditional models with filtered historical simulation pass in 4 of 4 (EWMA 12, GJR 13).

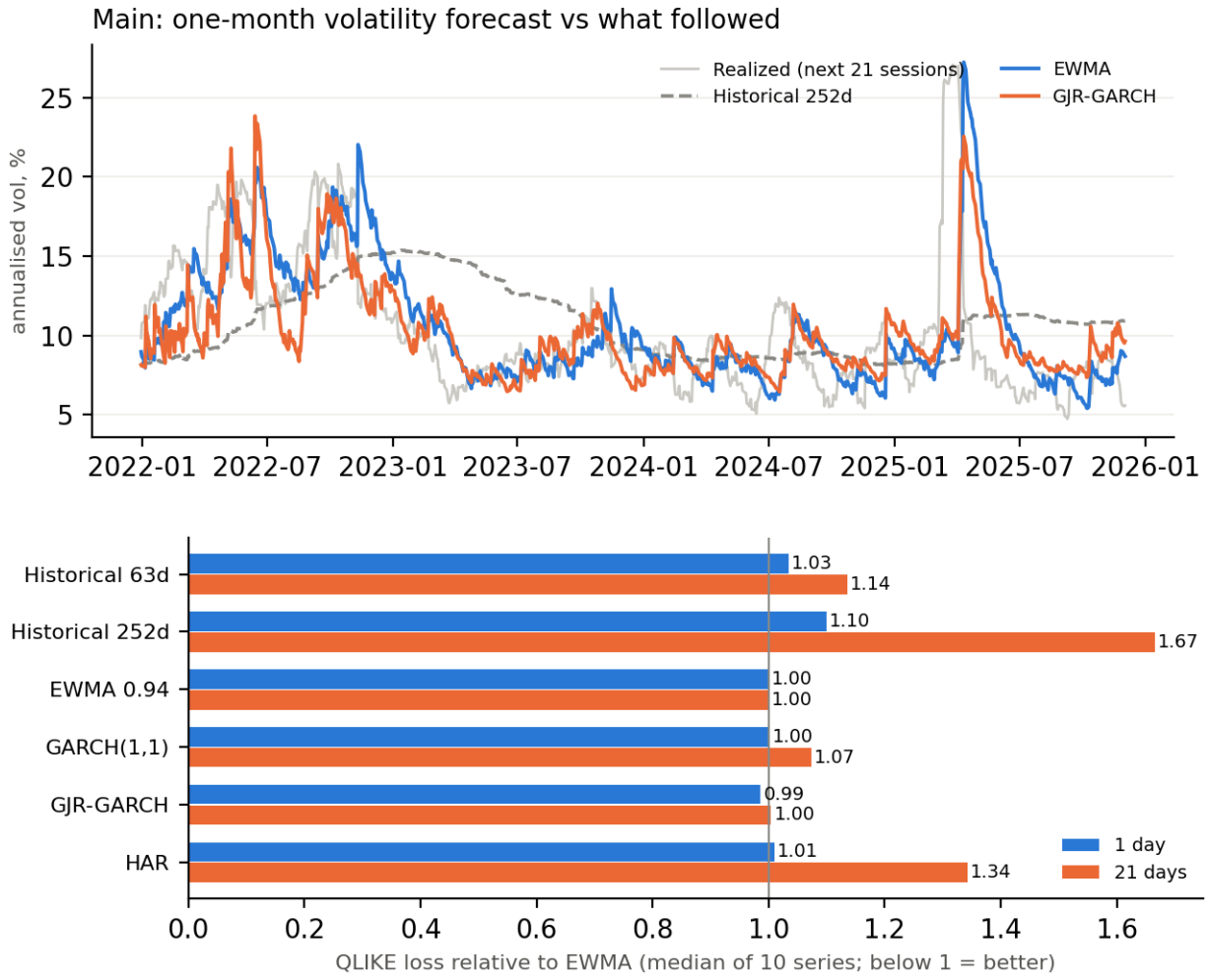
• One-month Mincer-Zarnowitz slopes are below 1 for every model (0.03 for the 252-day window, 0.64 for the best): no model is fully calibrated, and a year-long window says almost nothing about the next month. Four years containing two sharp episodes is a short sample for ranking models.

• Applied as a 10% volatility target without leverage, EWMA forecasts reduced Main's maximum drawdown from -21.4% to -17.2% with almost the same annual return (5.10% vs 5.16%); details on page 4.

Main portfolio: forecast accuracy

Model	Horizon	QLIKE	Rank	MZ slope	MZ R2	Avg forecast vol	Avg realized vol
GJR-GARCH	1d	1.469	1	0.96	0.080	10.19%	11.14%
EWMA 0.94	1d	1.501	2	0.71	0.049	10.49%	11.14%
GARCH(1,1)	1d	1.503	3	0.71	0.047	10.71%	11.14%
HAR	1d	1.507	4	0.81	0.046	10.80%	11.14%
Historical 63d	1d	1.555	5	0.59	0.024	10.59%	11.14%
Historical 252d	1d	1.701	6	0.22	0.001	10.78%	11.14%
GJR-GARCH	21d	0.215	1	0.64	0.192	10.23%	11.21%
EWMA 0.94	21d	0.238	2	0.43	0.137	10.55%	11.21%
HAR	21d	0.242	3	0.43	0.092	10.92%	11.21%
GARCH(1,1)	21d	0.243	4	0.39	0.104	10.92%	11.21%
Historical 63d	21d	0.273	5	0.43	0.095	10.65%	11.21%
Historical 252d	21d	0.394	6	0.03	0.000	10.78%	11.21%

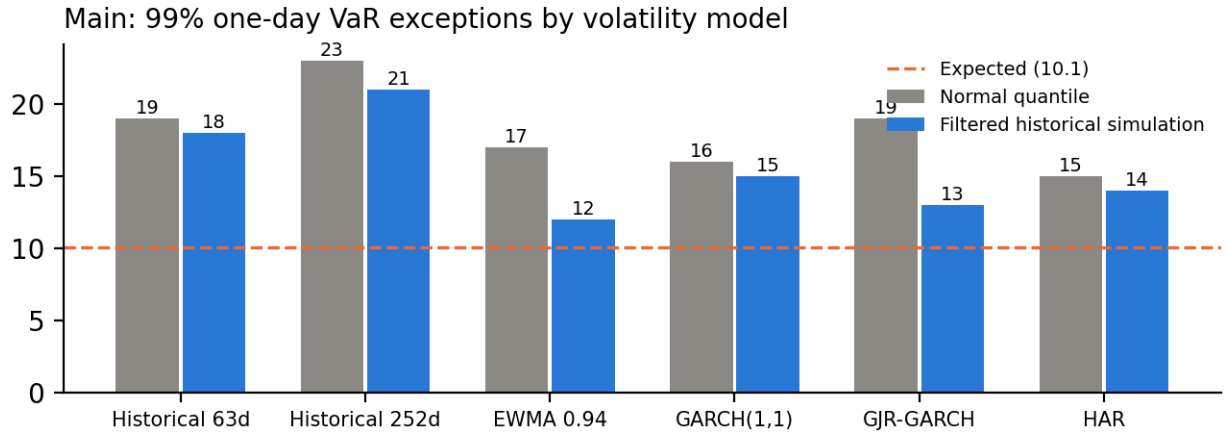
MZ slope = 1 and intercept = 0 indicate unbiased forecasts; slopes below 1 mean forecasts over-react. Squared daily returns are a very noisy target, so 1-day R2 values are low for every model.



Main portfolio: 1-day VaR backtest by volatility model

Model	Method	Conf.	Expected	Exceptions	Kupiec p	Indep. p	Avg VaR
EWMA 0.94	FHS	95%	50.3	54	0.592	0.953	1.12%
GARCH(1,1)	FHS	95%	50.3	54	0.592	0.555	1.12%
GJR-GARCH	FHS	95%	50.3	48	0.743	0.639	1.11%
HAR	FHS	95%	50.3	54	0.592	0.518	1.10%
Historical 252d	FHS	95%	50.3	57	0.338	0.140	1.18%
Historical 63d	FHS	95%	50.3	54	0.592	0.953	1.12%
EWMA 0.94	Normal	95%	50.3	50	0.971	0.736	1.09%
GARCH(1,1)	Normal	95%	50.3	47	0.635	0.590	1.11%
GJR-GARCH	Normal	95%	50.3	54	0.592	0.088	1.06%
HAR	Normal	95%	50.3	45	0.440	0.060	1.12%
Historical 252d	Normal	95%	50.3	50	0.971	0.043	1.12%
Historical 63d	Normal	95%	50.3	54	0.592	0.555	1.10%
EWMA 0.94	FHS	99%	10.1	12	0.549	0.130	1.83%
GARCH(1,1)	FHS	99%	10.1	15	0.144	0.217	1.84%
GJR-GARCH	FHS	99%	10.1	13	0.371	0.156	1.77%
HAR	FHS	99%	10.1	14	0.237	0.185	1.81%
Historical 252d	FHS	99%	10.1	21	0.002	0.456	1.90%
Historical 63d	FHS	99%	10.1	18	0.023	0.327	1.86%
EWMA 0.94	Normal	99%	10.1	17	0.045	0.288	1.54%
GARCH(1,1)	Normal	99%	10.1	16	0.082	0.251	1.57%
GJR-GARCH	Normal	99%	10.1	19	0.012	0.368	1.49%
HAR	Normal	99%	10.1	15	0.144	0.017	1.58%
Historical 252d	Normal	99%	10.1	23	0.000	0.108	1.58%
Historical 63d	Normal	99%	10.1	19	0.012	0.368	1.55%

Same 1005 days (2021-12-30 to 2025-12-31) for every method. Filtered historical simulation uses up to 250 past standardized returns; returns before the first forecast are standardized with information available at that date. $p < 0.05$ rejects correct coverage (Kupiec) or independence of exceptions.



Application: volatility targeting (risk control, not return forecasting)

Each month the risky exposure is set to min(100%, 10% / forecast one-month volatility); the rest earns the T-bill rate. Target, cap and a 5 bps cost on exposure changes were fixed in advance; 8% and 12% are sensitivity. Period 2022-01-03 to 2025-12-31. Sharpe ratios are in excess of the T-bill rate, which was high in 2023-2025, so they are low for every line.

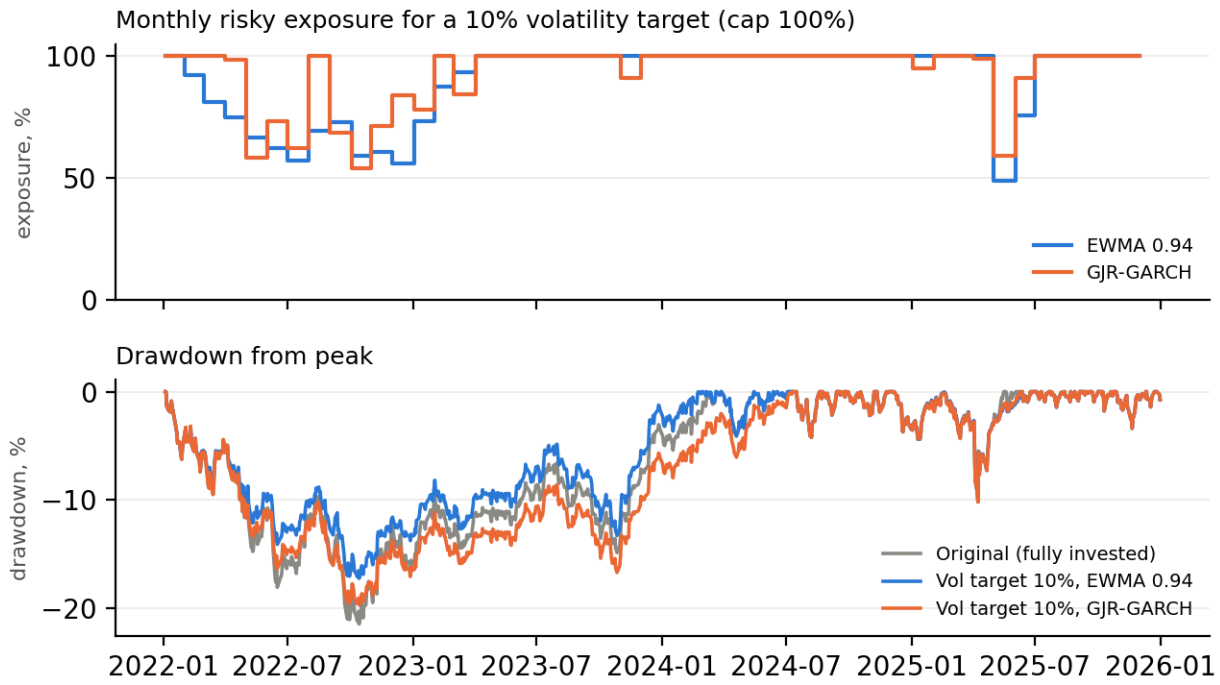
• With EWMA forecasts the overlay held 90% exposure on average and cut volatility from 11.2% to 9.5%, maximum drawdown from -21.4% to -17.2% and the worst month from -7.5% to -5.4%, while annual return moved from 5.16% to 5.10% (costs 11 bps in total). The Sharpe ratio is essentially unchanged (p = 1.00).

• With GJR-GARCH forecasts the same rule earned 4.06% a year with a smaller risk reduction (max drawdown -19.7%); its Sharpe ratio is lower but not significantly (p = 0.27). A model that forecasts slightly better on average did not make the better overlay here, a reminder that forecast accuracy and decision value differ.

• A monthly overlay cannot react to a shock inside the month: the highest monthly volatility stays at 28% (April 2025). Four years is far too short to claim the overlay adds value; the evidence is about controlling risk, and later research (Cederburg et al., 2020) finds volatility management rarely improves performance out of sample.

Strategy	Target	Avg exp.	Return/yr	Vol	Sharpe	Max DD	Worst mo.	ES 95%	Sharpe p
Original	-	100%	5.16%	11.15%	0.14	-21.4%	-7.5%	1.57%	-
EWMA 0.94	8%	82%	4.96%	8.45%	0.13	-14.4%	-4.3%	1.16%	0.94
EWMA 0.94	10%	90%	5.10%	9.53%	0.14	-17.2%	-5.4%	1.31%	1.00
EWMA 0.94	12%	94%	4.82%	10.16%	0.11	-19.7%	-6.5%	1.42%	0.68
GJR-GARCH	8%	84%	3.99%	8.70%	0.02	-17.0%	-5.0%	1.21%	0.33
GJR-GARCH	10%	93%	4.06%	9.93%	0.04	-19.7%	-6.3%	1.39%	0.27
GJR-GARCH	12%	96%	4.47%	10.45%	0.08	-20.8%	-6.3%	1.47%	0.32

Sharpe p: Jobson-Korkie test with Memmel's correction against the original. ES 95%: average loss on the worst 5% of days.



Verification and limitations

Unit tests recover known GARCH and GJR parameters from simulated data, confirm the optimum is a local minimum, check loss functions by hand and confirm no forecast changes when a later return is altered. On the real data, random perturbations of every fitted optimum raise the negative log-likelihood (smallest increase 7.08e-04). Limitations: four years of out-of-sample data, dominated by the 2022 rate shock and the April 2025 sell-off; squared returns are a noisy volatility proxy (intraday data would sharpen the comparison); Gaussian likelihood; the portfolio series is modelled directly rather than through an asset covariance model; nothing here forecasts the direction of returns.